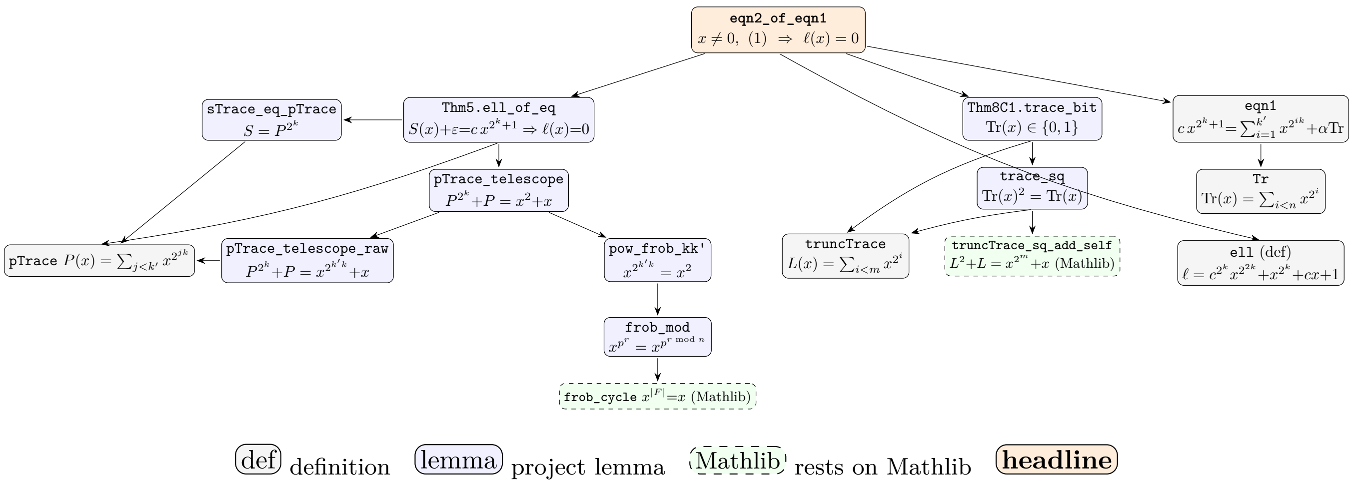
Dependency DAG of $(1) \Rightarrow (2)$

headline `eqn2_of_eqn1`: for $x \neq 0$, equation $(1) \Rightarrow \ell_k(c,x) = 0$ (edge $A \to B$: "A proved using B")



The three inputs and their Dobbertin source. The single step $(1) \Rightarrow (2)$ is the opening of the proof of Theorem 1 in Dobbertin, *Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets.* Verbatim (equations (1), (2)):

> *"…we shall show that for each fixed $c \in L$, the equation*
>
> $$c\,x^{2^k+1} = \sum_{i=1}^{k'} x^{2^{ik}} + \alpha\,\mathrm{Tr}(x) \tag{1}$$
>
> *has at most one solution in L. Adding the 2^k-th power of this equation to itself we get … and consequently*
>
> $$\ell(x) = c^{2^k} x^{2^{2k}} + x^{2^k} + cx + 1 = 0. \tag{2}$$
>
> *Note that $\ell(x) = 0$ if and only if $c\,x^{2^k+1} + \sum_{i=1}^{k'} x^{2^{ik}} + \alpha\,\mathrm{Tr}(x) = 0$ or 1."*

The Lean proof realises "adding the 2^k-th power to itself" by an Artin–Schreier telescoping, and the "$= 0$ or 1" by the fact that $\mathrm{Tr}(x)$ is a bit.

(A) Telescoping branch $\to$ `Thm5.ell_of_eq` (the real content of $(1) \Rightarrow (2)$):

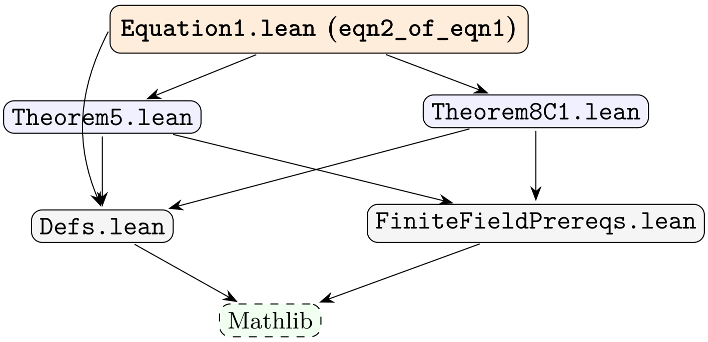
- `sTrace_eq_pTrace`: $S(x) = P(x)^{2^k}$, with $P(x) = \sum_{j=0}^{k'-1} x^{2^{jk}}$, $S(x) = \sum_{i=1}^{k'} x^{2^{ik}}$.
- `pTrace_telescope_raw`: $P(x)^{2^k} + P(x) = x^{2^{k'k}} + x$ (char 2, no field needed).
- `pow_frob_kk'`: if $kk' \equiv 1 \pmod{n}$ then $x^{2^{k'k}} = x^2$ (uses `frob_mod` $\to$ `frob_cycle`, i.e. $x^{|F|} = x$).
- `pTrace_telescope`: hence $P(x)^{2^k} + P(x) = x^2 + x$ (Artin–Schreier form).
- `bit_pow`: $\varepsilon \in \{0,1\} \Rightarrow \varepsilon^{2^m} = \varepsilon$.
- `ell_of_eq`: from $S(x) + \varepsilon = c\,x^{2^k+1}$ ($\varepsilon \in \{0,1\}$, $x \neq 0$), raising $P = (x^2+x) + c\,x^{2^k+1} + \varepsilon$ to the 2^k power and dividing by x^{2^k} gives $\ell(x) = 0$.

(B) Trace-parity branch $\to$ `Thm8C1.trace_bit` (this is the "$= 0$ or 1" above): $\mathrm{Tr}(x) \in \{0,1\}$.

- `truncTrace`: $L(x) = \sum_{i=0}^{m-1} x^{2^i}$ (the trace, def).
- `truncTrace_sq_add_self`: $L(x)^2 + L(x) = x^{2^m} + x$.
- `trace_sq`: on $\mathbb{F}_{2^n}$, $\mathrm{Tr}(x)^2 = \mathrm{Tr}(x)$ (uses $x^{|F|} = x$).
- `trace_bit`: an idempotent in a field is 0 or 1.

(C) Definitions branch (`Defs.lean`): `eqn1` is equation (1) cleared of denominators; `ell` is $\ell(x)$ of (2); $\texttt{Tr}(x) = \sum_{i<n} x^{2^i}$.

Module-level DAG.



Not on this path. `Theorem8Trace.lean`, `Q1General.lean` and the full `theorem_1`, `theorem_1_case1`, `theorem_1_case2` bijectivity machinery are *not* used by `eqn2_of_eqn1`: the single step (1) $\Rightarrow$ (2) needs only `Thm5.ell_of_eq` and `Thm8C1.trace_bit` plus the definitions.